

Finance Club, IIT Roorkee: Open Projects 2026

Project: Stochastic Interest Rate Modelling and Prediction

Implementing, Calibrating, and Extending the Cox-Ingersoll-Ross Model on Real Yield Curve Data



Project Type: Single-Person Project **Duration:** 2 Weeks **Eligibility:** Open to all students
Support: Mentorship by Finance Club members

1 Core Problem Statement

How can a stochastic short-rate model be implemented, calibrated against noisy historical yield data, and extended to reconstruct an entire yield curve from a single observable input—and where do such models succeed or fail when confronted with real market dynamics?

2 Project Overview

Interest rates are the fundamental building blocks of the global financial system. They dictate the pricing of bonds, the valuation of derivatives, and the risk management strategies of institutional portfolios. Yet interest rates are not static; they evolve over time in a complex, seemingly random manner that resists simple forecasting. To capture this evolution, quantitative analysts rely on advanced mathematical frameworks rooted in stochastic calculus.

In this project, you will dive into the world of short-rate modelling. Your primary objective is to implement, calibrate, and improve upon the Cox-Ingersoll-Ross (CIR) model. Unlike simpler models, the CIR framework uses a mean-reverting square-root diffusion process to ensure that interest rates remain strictly positive—a crucial characteristic for real-world macroeconomic modelling. You will build this model from the ground up, estimate its parameters from historical market data, project the yield curve, and critically analyse its mathematical and practical limitations.

3 Background and Introduction

The CIR model, introduced by Cox, Ingersoll, and Ross in 1985, describes the evolution of the instantaneous short rate r_t via the stochastic differential equation:

$$dr_t = \kappa(\theta - r_t) dt + \sigma\sqrt{r_t} dW_t$$

where $\kappa > 0$ is the speed of mean reversion, $\theta > 0$ is the long-run mean, $\sigma > 0$ is the volatility coefficient, and W_t is a standard Brownian motion. The square-root diffusion ensures that rates cannot become negative provided the Feller condition $2\kappa\theta \geq \sigma^2$ is satisfied.

From the short rate, the model derives a closed-form bond pricing formula. The price at time t of a zero-coupon bond maturing at T is given by:

$$P(t, T) = A(t, T) e^{-B(t, T) r_t}$$

where $A(t, T)$ and $B(t, T)$ are deterministic functions of the model parameters and time to maturity $\tau = T - t$. The continuously compounded yield for maturity τ is then:

$$y(t, \tau) = -\frac{\ln P(t, T)}{\tau} = \frac{B(t, T) r_t - \ln A(t, T)}{\tau}$$

This closed-form tractability is what makes the CIR model both practically useful and a natural starting point for extensions.

While foundational, the base CIR model carries well-documented limitations. It cannot perfectly fit an arbitrary initial term structure, struggles to capture sudden market shocks, and constrains the yield curve to shapes that a single factor can produce. These limitations motivate the advanced extensions you will explore.

4 The Dataset

You will be provided with a historical dataset containing daily bond yields.

- **The Underlying Asset:** The specific market, geographical region, or asset class from which these yields are derived will remain undisclosed. Your model must rely purely on the mathematical relationships within the data, without macroeconomic bias.
- **Maturities:** The dataset includes yield data across 9 specific maturity tenors: 3 Months, 6 Months, 9 Months, 1 Year, 2 Years, 5 Years, 10 Years, 20 Years, and 30 Years.
- **Data Quality:** The provided CSV files will not be clean. You will encounter missing values, outliers, formatting inconsistencies, and non-trading day anomalies. Robust data preprocessing is a mandatory first step.
- **Splitting:** You will receive both a Training Dataset and a Full Test Dataset to facilitate out-of-sample backtesting.

Dataset Link: [Google Drive](#)

5 Core Objectives & Workflow

Your project submission must chronologically address the following milestones.

5.1 A. Data Engineering and Preprocessing

Clean the daily time-series data. Handle missing values appropriately—for example through interpolation or forward-filling—detect and normalise outliers, and ensure the data is mathematically viable for time-series calibration.

5.2 B. Base CIR Model Implementation & Calibration

Implement the mathematical framework to simulate and calibrate the CIR process. You are free to use any calibration methodology you deem appropriate to find the optimal parameters (κ, θ, σ) . Common approaches include Ordinary Least Squares (OLS), Maximum Likelihood Estimation (MLE), Generalised Method of Moments (GMM), or advanced techniques such as Kalman Filtering. You must justify your chosen method.

5.3 C. The Prediction Challenge: Yield Curve Construction

This is the core test of your model’s predictive power. When generating your predicted yield curve (6M through 30Y) for any given day in the test period, your prediction algorithm is **only permitted to ingest the 3-Month (3M) yield** for that day as a proxy for the instantaneous short rate r_t . You will use your calibrated parameters and this 3M rate to theoretically reconstruct the entire yield curve and compare it against the held-out actuals in the test set.

5.4 D. Model Improvement & Extensions

The base CIR model, while foundational, has well-documented limitations. You are required to research, implement, and backtest at least one advanced extension. Potential avenues include:

- **Two-Factor CIR Models:** Introducing a second stochastic factor to capture different variations in the yield curve such as level versus slope. The Longstaff-Schwartz (1992) model is a canonical reference.
- **Jump-Diffusion Processes:** Incorporating Poisson jump processes into the SDE to account for sudden macroeconomic shocks or central bank policy announcements, following the framework of Duffie, Pan, and Singleton (2000).
- **Time-Dependent Parameters (CIR++):** Allowing parameters to be deterministic functions of time, as developed by Brigo and Mercurio, so that the initial yield curve is fitted exactly.

5.5 E. Critical Analysis

Identify and articulate the practical and theoretical limitations of both your base model and your extended model. Discuss the implications of your mathematical assumptions in a real-world trading or risk-management scenario.

6 Key Questions the Project Must Answer

6.1 Model mechanics and calibration

- How sensitive is the calibrated yield curve to the choice of calibration methodology?
- Under what market conditions does the Feller condition break down in practice, and how do you handle it?
- What does the mean-reversion speed κ imply about the persistence of interest rate shocks in your data?

6.2 Prediction and out-of-sample performance

- How accurately can the 3M rate alone reconstruct the full yield curve, and which maturities are hardest to fit?
- Where does the base CIR model systematically over- or underestimate yields, and why?
- Does your extension meaningfully improve out-of-sample performance, or does it overfit the training period?

6.3 Extensions and modelling choices

- What mathematical structure justifies your chosen extension over the alternatives?
- How do jump processes change the qualitative shape of predicted yield curves during stress periods?
- What are the additional estimation challenges introduced by a two-factor or time-dependent model?

7 Deliverables & Formatting Requirements

This project requires the submission of a single Google Colab Notebook. No separate PDF reports, slide decks, or local Python scripts will be accepted.

Requirement	Description
Code Execution	The notebook must run fully from top to bottom without errors, fetching and processing data, calibrating models, and outputting all evaluation metrics.
Markdown as Report	Markdown cells integrated between code blocks must explain the stochastic calculus, justify calibration choices, detail extensions, and outline limitations.
Code Quality	Write Pythonic, modular, and readable code using functions and classes where appropriate, with clear inline comments on complex mathematical steps.
Access	Set the Google Colab link sharing to “Anyone with the link can view”.

8 Evaluation Criteria

Only submissions meeting **both** of the following criteria will be verified by the Finance Club.

1. **Predictive Accuracy (Out-of-Sample R^2):** The out-of-sample R^2 score achieved on the test dataset when reconstructing the yield curve from the 3M rate must be **greater than 0.85**.
2. **Readability and Code Quality:** Code must be Pythonic, modular, and well-commented, using functions and classes where appropriate with clear inline comments explaining complex mathematical transformations.

If you have any queries regarding the problem statement, please reach out to the Finance Club. Good luck!